



ENGINEER & MARVEX IDEAS COMPETITION: PITCH FOR TOMORROW

THEME: The Responsive Skin
FOCUS: Passive Cooling via Kinetic Design

TOPIC:

Snowflake Kinetic Louvre Facade

Number of subtopic: ____1____

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1.0 Executive Summary

The Problem

Static facades in tropical commercial buildings are a liability, not a feature. HVAC already consumes up to 70% of commercial building energy in Malaysia with space cooling driving 42–49% of total electricity use and most facades sit right at the regulatory OTTV limit, with almost no buffer left as heat loads rise. Static shading can't track the sun, forcing a constant trade-off between daylight and cooling efficiency.

The Solution: Snowflake Kinetic Louvre Facade

A biomimetic, modular, sensor-driven facade inspired by a snowflake's six-fold radial geometry. Each module pairs a **fixed structural frame** with **rotating trapezoidal Louvres** (0°–45°) that track the sun in real time, filtering direct heat while letting diffuse daylight through. LDR sensors and an Arduino controller drive servo-actuated Louvres via a synchronized linkage with rotary-encoder feedback closing the loop.

Why It's Different

Unlike Al Bahar Towers or CH2, which move entire structural panels, only the lightweight internal Louvres rotate while the frame stays fixed. This cuts mechanical complexity, wind vulnerability and maintenance cost, while staying retrofit-compatible with existing curtain walls.

Grounded in Research

Al Bahar Towers proves kinetic shading works (>50% solar gain reduction). Menara Mesiniaga proves Malaysia already values climate-tuned shading but only in static form. Our system addresses this gap by combining modular kinetic shading with a climate-responsive design suited to tropical commercial buildings.

The Numbers

Target: ≥25% cooling-load reduction, subject to thermal and building-energy validation. RM 1.5276M initial investment (750 modules, 3,000 m²), RM182,882/year savings, ~8.35 years payback (post-GITA), 11.97% annual ROI, 25+ year service life, RM482,400 GITA tax saving..

Broader Impact

Improves occupant comfort and daylight quality, creates demand for facade engineering, fabrication and automation skills, aligns with Malaysia's NETR and qualifies for GITA/GTFS incentives and advances SDG 7, 9, 11 and 13.

Bottom Line

The Snowflake Kinetic Louvre Facade turns a proven biomimetic principle of motion localized to the smallest necessary part into a commercially viable, retrofit-ready facade that cuts cooling costs, preserves daylight and pays for itself within its lifespan.

2.0 Problem Statement

Problem Statement:

Static building facades in tropical climates can contribute significantly to solar heat gain in tropical commercial buildings, forcing HVAC systems to operate at peak capacity.

The Challenge:

Design a modular, sensor-driven kinetic facade system for a commercial high-rise that autonomously adjusts its shading angle based on real-time solar tracking. Teams should propose and justify how their adaptive facade strategy could contribute toward a targeted 25% reduction in cooling loads while maintaining effective natural indoor daylighting and occupant comfort.

2.1 Introduction

Malaysia's equatorial climate exposes glazed office facades to intense solar radiation and static shading forces a trade-off: over-shading loses daylight, under-shading pushes HVAC to peak capacity. This stems from shading built for average sun position rather than real-time conditions, costly retrofits and wind-vulnerable kinetic systems. Our solution is a bio-mimetic kinetic mechanism modelled on a snowflake, splitting two functions: a rigid hexagonal frame for structural stability and responsive Louvres nested within it that rotate to control solar intensity.

2.2 Objective

To develop the Snowflake Kinetic Louvre Facade, a modular, bio-mimetic system that autonomously adjusts shading angle in real time, cutting cooling loads by at least 25% while preserving daylighting without complex and high-maintenance mechanisms.

Value Framework

Dimension	Target Outcome
Environmental	Reduced solar heat gain and HVAC reliance, lowering the building's carbon footprint
Monetary Savings	Lower cooling costs; modular parts keep installation and maintenance spend low
Operational Efficiency	Low-friction rotation reduces actuator wear, extending service life
Occupant Well-being	Consistent daylight and outdoor visual connection, less glare and artificial lighting
Scalability	Independent modules allow phased, retrofit-friendly installation and easy replacement

3.0 Research & Literature Review

3.1 Tropical Countries with High Cooling Demand

Cooling demand is rising fastest in tropical economies, with over 80% of projected cooling electricity growth by 2050 occurring there and Southeast Asia's AC stock set to grow ninefold by 2040 (IEA, 2025). As an equatorial country, Malaysia sits within this high-growth zone, where cooling is a year-round baseline rather than a seasonal load.

3.2 Kinetic Facade Systems & Smart Shading Technologies (Global)

Several global buildings prove mechanically responsive shading can cut solar heat gain without sacrificing daylight. Al Bahar Towers (Abu Dhabi) uses a mashrabiya-inspired kinetic screen reducing solar gain by over 50% (Architecture Lab, 2025) while Council House 2 (Melbourne) and Q1 Headquarters (Germany) use comparable sun-tracking louvre systems. All three rely on large-scale motorized arrays, effective but costly to scale down into a modular unit.

3.3 Passive Cooling & Daylighting Balance Principles

Adaptive shading must balance heat rejection against daylight: over-shading raises lighting load, under-shading raises cooling load. Our solution resolves this by decoupling a static structural frame from an independently adjustable, sun-tracking Louvre layer.

3.4 Case Studies in Similar Tropical Climates

Locally, Menara Mesiniaga (Subang Jaya) demonstrates shading tuned to Southeast Asia's climate through external louvres, but remains static and fixed (Douglass-Jaimes, 2018). This reveals two disconnected strands: large-scale kinetic systems abroad (Section 3.2) prove adaptive shading works but rely on costly motorized structural panels while local precedent shows static climate-responsive shading but no real-time adaptability. No modular, small-scale kinetic facade combining both was identified in the literature reviewed — the gap our proposal targets.

3.5 Data & Statistics: Cooling Load and HVAC Energy Consumption

HVAC accounts for up to 70% of energy use in Malaysia's commercial sector. (Samuji et al., 2026). High-rise office buildings in Malaysia use approximately 45.9% of their total building energy for cooling purposes (Lau et al., 2016). The Energy Code caps OTTV at 50%, yet typical facade audits in this range record values approaching the limit, leaving most facades with little buffer for climate volatility.

3.6 Related SDGs

SDG 7 – Affordable and Clean Energy	Reduced cooling-driven electricity demand.
SDG 9 – Industry, Innovation and Infrastructure	Applied bio-mimetic innovation in building infrastructure.
SDG 11 – Sustainable Cities and Communities	Scalable, retrofit-friendly shading.
SDG 13 – Climate Action	Lower HVAC-driven emissions.

4.0 Proposed Solutions

4.1 Concept Overview

Adaptive Snowflake Kinetic Facade

The proposed solution is a **biomimetic, modular and sensor-driven kinetic facade** designed to reduce solar heat gain in commercial high-rise buildings located in tropical climates. Inspired by the **six-fold radial modular geometry of natural snowflakes**, the facade combines a fixed structural frame with an adaptive internal louvre system to provide intelligent solar control.

The six-fold radial symmetry of natural snowflakes forms a **repeatable hexagonal module** that distributes shading more uniformly across the facade while enabling efficient modular assembly, scalability and architectural flexibility for high-rise applications.

Each facade module consists of two primary components:

- **A fixed snowflake-shaped outer frame** which serves as the main structural support and acts as a protective wind-break for the internal kinetic louvre system and is fitted with thin-film photovoltaic strips to power the module's sensing and actuation system.
- **An adjustable trapezoidal louvre assembly**, consisting of radially arranged trapezoidal louvre blades. The blades gradually increase in length and width from the centre towards the outer edge, following the expanding geometry of each snowflake sector. This tapered arrangement preserves the snowflake-inspired appearance while providing more uniform solar shading across the facade.

Unlike conventional parallel rectangular louvres, the proposed tapered trapezoidal blades follow the radial geometry of the snowflake. This arrangement preserves the biomimetic appearance while maintaining consistent louvre rotation and effective solar shading. The varying blade width enables the shading system to integrate seamlessly within the snowflake frame without leaving large unused spaces.

Rather than completely blocking sunlight, the adjustable louvres selectively filter direct solar radiation while allowing diffuse daylight to pass through the spaces between the blades. This balances thermal comfort, indoor visual comfort and energy efficiency by reducing the demand for both air-conditioning and artificial lighting.

See Appendix A for AI-assisted visualizations of the facade in both operating states.

4.2 Working Principles

System Operation

The proposed kinetic facade consists of a **fixed snowflake-shaped outer frame** and an **internal adjustable louvre system**. The snowflake frame provides structural support and acts as a protective wind-break, while the louvres dynamically adjust their shading angle to regulate solar heat gain and maintain optimal indoor daylight.

The facade operates according to solar conditions:

(a) Low solar intensity

- Louvres remain at **0° (fully open)**.
- Maximum natural daylight enters the building.
- Minimal shading is required.

(b) High solar intensity

- Louvres rotate progressively up to **45°**.
- Direct solar radiation is reduced.
- Diffuse daylight continues to enter through the gaps between the louvre blades.
- Cooling loads are reduced while maintaining visual comfort.

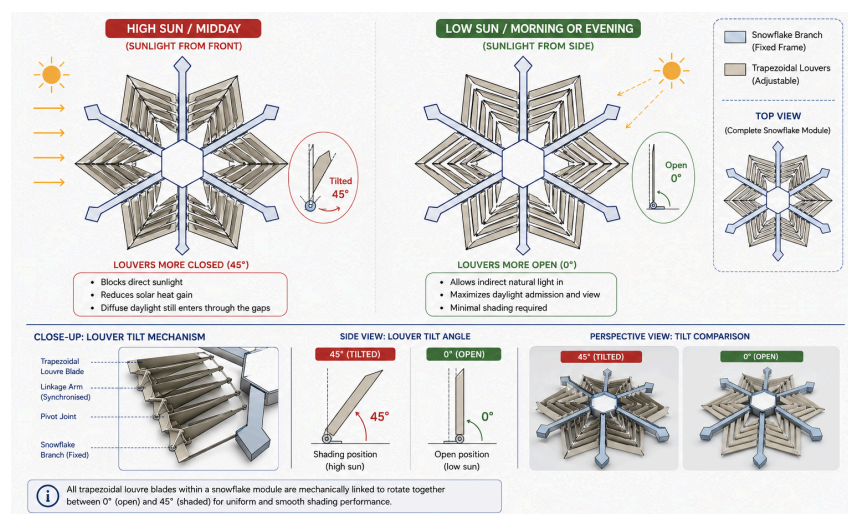


Figure 4.2.1 Louvre Movement during High and Low Sunlight Intensity

Operational Mechanism & Control System

The adaptive control system operates as a closed-loop process:

(a) Sensing

(i) Prototype:

An array of Light Dependent Resistors (LDRs) detects sunlight intensity. For the prototype, LDR sensors provide real-time light input to the controller.

(ii) Full Scale System:

The LDR-based sensing approach can be enhanced with a Solar Position Algorithm (SPA). By combining the building GPS coordinates with the current date and time, a

built-in mathematical formula can calculate the absolute position of the sun (azimuth) and elevation angles. These values will then be sent to the facade control system.

(iii) The LDR sensors, Arduino controller and servo motors are powered by photovoltaic strips integrated into the fixed frame, reducing dependence on external grid supply for the control system.

(b) Decision Making

For prototype, the Arduino microcontroller analyses the sensor data and determines the appropriate louvre angle before sending control signals to the servo motors. In the full-scale system, the control logic can incorporate both real-time sensor feedback and solar-position data to determine the required shading angle.

(c) Actuation

For the prototype, the servo motors receive the control signal and rotate the louvre mechanism through a linkage system. For full-scale implementation, appropriately rated actuators would be used to provide the required torque and durability for the facade.

(d) Motion Transmission

The mechanical linkage synchronises the rotation of all trapezoidal louvre blades within each snowflake module, ensuring smooth and uniform movement. Adequate spacing between adjacent blades prevents mechanical interference throughout the 0° – 45° rotation range, maintaining consistent shading performance.

(e) Position Feedback

For full scale implementation, the Magnetic Absolute Rotary Encoder monitors the real time angles of the louvre blades by using a magnetic field. It tells the motor to stop the moment the encoder detects that the louvre has reached the shading angle needed.

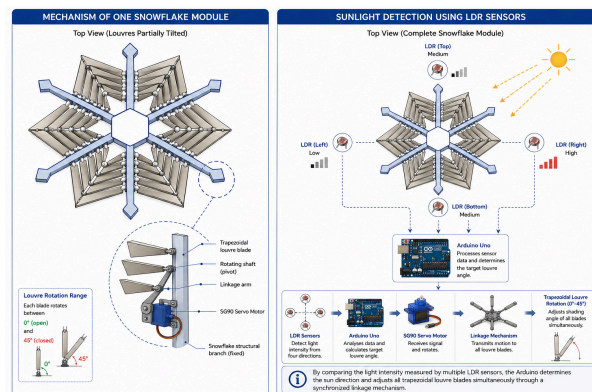
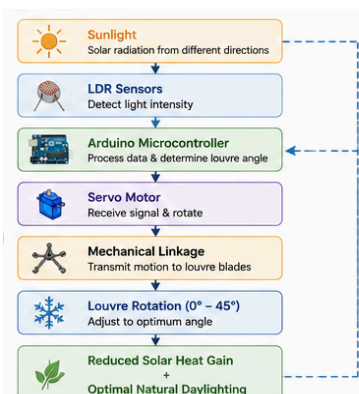


Figure 4.2.2 Mechanism of a Snowflake Module and Sunlight Detection using LDR Sensors

Control Flow



Design Principles

Modular Design

Each snowflake functions as an independent facade module that can be manufactured, installed and replaced individually. This modular configuration also enables the facade to be easily expanded across buildings of different sizes and orientations.

Design for Disassembly

Mechanical fasteners are used to connect the frame, louvres, sensors and actuators, allowing damaged components to be removed and replaced without dismantling the entire facade.

Passive Cooling

The adjustable louvres reduce direct solar heat gain before it reaches the building envelope, lowering indoor cooling demand while maintaining comfortable indoor conditions.

Adaptive Daylighting

Instead of completely blocking sunlight, the louvres regulate the amount of incoming daylight through the rotation of tapered trapezoidal louvre blades arranged radially within each snowflake module.

Geometric Feasibility Verification

A simplified single-module geometric layout was developed to verify mechanical feasibility of the trapezoidal louvre arrangement. Assuming a hexagonal snowflake frame with a radius of 1.5 m, six radial branches each host 4 tapered trapezoidal louvre blades increasing from 80 mm (innermost) to 220 mm (outermost) in width (nominal blade width, measured perpendicular to the blade edge) with a uniform 15 mm inter-blade gap maintained across the full 0°–45° rotation range to prevent mechanical interference. This spacing was verified through basic trigonometric clearance calculation: at 45° tilt, blade overlap clearance = blade width $\times \sin(45^\circ) \approx 0.71 \times$ blade width, confirming sufficient gap retention for the 220 mm maximum blade width within the 1.5 m module radius. Full dimensional workings are provided in Appendix D.

4.3 Feasibility, Safety and Compliance Considerations

Feasibility

(a) Cost

The prototype uses affordable components — Arduino Uno, LDR sensors and SG90 servo motors — to demonstrate the core mechanism. Full-scale implementation would use appropriately rated components, while integrated PV strips reduce reliance on grid electricity.

(b) Maintenance

The modular configuration lets individual units, sensors or servo motors be serviced without dismantling the facade and the simple rotational mechanism keeps maintenance low. The frame-mounted photovoltaic strips have no moving parts and need only periodic cleaning, largely handled by Malaysia's regular rainfall.

(c) Retrofitting

The proposed snowflake modules are designed as secondary facade units that can be retrofitted onto existing commercial curtain wall systems with minimal structural

modification. This allows existing buildings to improve energy performance without replacing the primary building envelope.

Engineering Design Principles

The proposed facade applies several engineering principles that improve both performance and practicality.

- (a) **Modularity:** Independent snowflake modules simplify manufacturing, transportation, installation and future replacement while allowing the system to be scaled for different building sizes.
- (b) **Sensor-Driven Automation:** LDR sensors provide continuous real-time light measurements, enabling the Arduino-based controller to automatically adjust the louvre angle without manual intervention.
- (c) **Mechanical Simplicity:** The rotational louvre mechanism uses fewer moving components than folding or telescopic facade systems, reducing mechanical wear, energy consumption and maintenance requirements.
- (d) **Passive Environmental Control:** By reducing solar heat gain before it enters the building envelope, the adaptive louvre system decreases HVAC cooling demand while maintaining indoor thermal comfort.
- (e) **Future Scalability:** Although the prototype relies on LDR sensors, a full-scale implementation could combine real-time sensor feedback with a pre-programmed **solar position (solar vector) algorithm**, which calculates the sun's azimuth and altitude based on the building's location, date and time. This would enable more accurate and predictive louvre adjustment throughout the day.

Preliminary Cooling-Load Reduction Estimation

A preliminary solar heat-gain analysis was conducted to assess the potential of the proposed Snowflake Kinetic Louvre Facade to contribute towards the targeted 25% cooling-load reduction. The calculation serves as an initial engineering feasibility assessment rather than a validated whole-building energy simulation. The analysis applies the Solar Heat Gain Coefficient (SHGC) method to the 3,000 m² protected glazing area. A representative SHGC value of 0.40 and a peak façade solar irradiance of 700 W/m² were used, with the detailed calculations provided in Appendix C.

At the maximum louvre rotation angle of 45°, the proposed louvre system is preliminarily estimated to intercept approximately 80% of direct solar radiation while allowing approximately 70% of diffuse radiation to pass through the gaps between the louvre blades. The 80% value is informed by findings that effective external shading can substantially reduce solar heat gain through glazed façades (Palmero-Marrero & Oliveira, 2010).

Based on the stated assumptions, the estimated solar heat gain is reduced from approximately **840 kW** to **315 kW**, representing a **62.5% reduction in solar heat gain**.

As solar heat gain is only one component of the total cooling load, it is preliminarily assumed to contribute approximately 40% of the cooling load within the protected highly glazed perimeter zones. Therefore:

Estimated Cooling-Load Reduction

= Solar Heat-Gain Reduction × Solar Contribution to Cooling Load

$$= 62.5\% \times 40\%$$

$$= \mathbf{25.0\%}$$

Based on these preliminary assumptions, the proposed Snowflake Kinetic Louvre Facade demonstrates the potential to contribute towards the targeted 25% cooling-load reduction within the protected glazed perimeter zones. The actual cooling-load reduction will depend on facade orientation, solar conditions, building operation and control performance and will require validation through prototype testing and thermal or building-energy simulation. The complete calculation is provided in Appendix C.

Safety and Compliance Considerations

The proposed design incorporates several safety and operational considerations:

- Aluminium and stainless steel components provide high structural strength, corrosion resistance and long-term outdoor durability.
- Electrical components, including servo motors and wiring, are enclosed within the snowflake frame to protect them from dust, moisture and accidental contact.
- Rounded frame edges minimise injury risks during installation and maintenance.
- A manual override function can be incorporated to allow operation during electrical or control system failure.
- For future large-scale applications, a **wind safety mode** may be integrated, allowing the louvres to automatically return to the fully open (0°) position during extreme wind conditions to reduce wind loads on the movable components.
- The proposed materials are also compatible with commonly used architectural facade systems and are suitable for meeting standard building durability and fire safety requirements.

4.4 Unique Value Proposition

Unique Value Proposition

Unlike conventional kinetic facades that rely on rotating structural panels or flower-inspired mechanisms, the proposed **Snowflake Kinetic Louvre Facade** combines **biomimetic architecture**, **adaptive solar control** and **modular engineering**, with frame-integrated photovoltaic strips powering the sensing and actuation system.

Rather than moving the entire facade panel, only the lightweight internal louvres rotate while the snowflake frame remains fixed. This significantly reduces mechanical complexity while preserving the distinctive snowflake-inspired architectural appearance.

The system provides intelligent, sensor-driven shading that reduces solar heat gain while maintaining optimal natural daylight, contributing to improved energy efficiency and occupant comfort.

Key Innovations

- **Snowflake-inspired biomimetic facade geometry**, creating a unique architectural identity.
- **Static structural frame with dynamic internal louvres**, reducing moving mass and improving reliability.

- **Real-time sensor-driven louvre adjustment** for adaptive solar control.
- **Frame-integrated photovoltaic strips**, reducing the system's dependence on grid electricity for sensing and actuation.
- **Passive cooling** that lowers HVAC energy demand while maintaining indoor daylight quality.
- **Modular design for disassembly**, allowing easier maintenance, replacement and future expansion.
- **Retrofittable facade modules** suitable for both new and existing commercial buildings.

Comparison with Existing Solutions

Feature	Conventional Kinetic Facades	Static Louvres	Proposed Snowflake Kinetic Louvre Facade
Mechanical Complexity	High (moving structural panels)	Very Low	Low–Medium (static frame with rotating internal louvres)
Solar Adaptability	Dynamic tracking	Fixed shading angle	Real-time sensor-driven adaptive shading
Daylighting Performance	May reduce daylight when fully closed	Limited adaptability	Maintains optimal daylight while reducing heat gain
Passive Cooling Performance	Good	Moderate	High, through adaptive louvre control
Maintenance	Complex moving mechanisms	Low maintenance	Modular replaceable units with simple rotational mechanism
Retrofitting	Often difficult	Moderate	Suitable for retrofit onto existing curtain wall systems
Architectural Identity	Mostly conventional geometric panels	Conventional	Distinctive snowflake-inspired façade with tapered radial louvre geometry

The proposed design differentiates itself from conventional biomimetic facades by using the snowflake primarily as a modular architectural and structural framework, while integrating an adaptive kinetic louvre assembly to achieve intelligent solar control. This combination delivers a facade that is visually distinctive, mechanically practical and scalable for real-world commercial buildings.

5.0 Social, Economy and Environmental Impact Analysis

5.1 Social, Industrial and Community Impact

5.1.1 Social Impact

The proposed Snowflake Kinetic Louvre Facade improves indoor environmental quality in tropical commercial buildings by reducing solar heat gain, glare and excessive cooling demand. Through adaptive louvre adjustment, the system balances solar control and daylight penetration, enhancing thermal comfort, visual comfort and occupant well-being.

5.1.2 Industrial Impact

The proposed façade integrates façade engineering, aluminium fabrication, mechanical systems and building automation, supporting innovation in smart-building technologies. The modular design enables prefabrication, easier installation and component replacement, while creating opportunities in façade manufacturing, automation control, engineering services and maintenance.

5.1.3 Community Impact

The proposed system contributes to sustainable urban development by reducing building cooling demand and operational carbon emissions. Its retrofit capability allows improvement of existing curtain-wall buildings without replacing the primary envelope. The snowflake-inspired design also promotes awareness and adoption of biomimetic and climate-responsive architecture.

Overall, the Snowflake Kinetic Louvre Facade provides social benefits through improved occupant comfort, industrial benefits through smart façade innovation and community benefits through sustainable building development.

5.2 Economic and Financial Assessment

5.2.1 Preliminary Cost Estimation

The proposed building has a floor area of 10,000 m², with approximately 4,000 m² of glazing. The kinetic façade covers approximately 3,000 m², requiring 750 snowflake modules of 2 m × 2 m.

The complete proposed façade integrates photovoltaic strips directly into the fixed branches of each snowflake module.

The proposed façade may qualify for the Green Investment Tax Allowance (GITA), subject to the applicable eligibility requirements and approval. Based on Malaysia's corporate income tax rate of **24%**, the potential tax saving is calculated as:

$$\begin{aligned}\text{GITA Tax Saving} &= \text{Gross Initial Investment} \times \text{Corporate Tax Rate} \\ &= \text{RM2,010,000} \times 24\% \\ &= \text{RM482,400}\end{aligned}$$

$$\begin{aligned}\text{Effective Net Investment} &= \text{Gross Initial Investment} - \text{GITA Tax Saving} \\ &= \text{RM2,010,000} - \text{RM482,400} \\ &= \text{RM1,527,600}\end{aligned}$$

Preliminary Investment Estimation

Cost Component	Estimated Cost (RM)
Aluminium snowflake frame and louvre blades	1,100,000
Stainless steel mechanical supports and linkage	250,000
LDR sensors and Arduino-based control system	150,000
Servo motors and actuator system	200,000
Electrical wiring and protective housing	100,000
Installation and commissioning	100,000
PV strips (10 W/module × 750 modules)	30,000
PV power-management and installation	30,000
Contingency and engineering cost	50,000
Gross Initial Investment	2,010,000
GITA tax saving at 24%	(482,400)
Effective Net Investment	1,527,600

Although the initial cost is higher than conventional static shading systems, the adaptive capability provides additional value through reduced cooling demand, improved daylight management and enhanced occupant comfort.

5.2.2 Energy Saving Estimation and Potential Savings

For preliminary economic assessment, a 25% cooling-demand reduction is adopted as the design target based on the preliminary solar heat-gain analysis in Section 4.3. This value is not a validated whole-building performance result and will be refined following prototype testing and building-energy simulation.

Estimated Annual Financial Benefits

Benefit Category	Estimated Annual Saving
HVAC cooling-energy reduction	RM108,000
Reduced artificial lighting demand	RM24,000
Reduced maintenance and replacement cost	RM50,000
PV electricity cost saving	RM 882
Total Annual Benefit	RM182,882/year

5.2.3 Payback Period and Return on Investment (ROI)

The simple payback period is calculated based on the initial investment and annual financial benefits.

Payback Period = Initial Investment ÷ Annual Financial Benefit

= RM1,527,600 ÷ RM182,882/year

= 8.35 years

Annual ROI = (Annual Financial Benefit ÷ Initial Investment) × 100%

= (RM182,882 ÷ RM1,527,600) × 100%

= 11.97% per year

Economic Performance Summary

Economic Indicator	Estimated Value
Initial investment	RM1,527,600
Annual financial benefit	RM182,882/year
Payback period	8.35 years
Annual ROI	11.97%
Expected service life	>25 years

The estimated payback period demonstrates the long-term economic feasibility of the proposed façade. Although the system requires higher initial investment, the operational savings throughout its service life provide additional financial value.

5.2.4 Material Sourcing Cost Comparison and Selection

Material selection influences the initial cost, durability and sustainability performance of the kinetic façade. Recycled materials were considered as alternatives to conventional materials to reduce material cost and support circular economy principles.

Aluminium components were prioritised for recycled material application because aluminium can be repeatedly recycled while maintaining its mechanical properties.

Material Sourcing Cost Comparison (Conventional vs Recycled Materials)

Component	Conventional Material	Estimated Cost	Recycled Alternative	Estimated Cost
Snowflake frame	Virgin aluminium alloy extrusion	RM12–15/kg	Recycled aluminium alloy extrusion	RM9–12/kg
Louvre blades	Virgin aluminium sheet	RM10–14/kg	Recycled aluminium sheet	RM8–11/kg
Mechanical linkage	Stainless Steel 316	RM15–25/kg	Recycled Stainless Steel 316	RM13–22/kg

Electrical housing	Virgin polycarbonate	RM20–30/kg	Recycled polycarbonate	RM15–25/kg
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Based on the comparison, recycled aluminium alloy is selected for the snowflake frame and louvre blades due to its lightweight properties, corrosion resistance, recyclability and cost advantage. Stainless Steel 316 remains selected for mechanical components because high corrosion resistance is required for long-term outdoor operation.

Final Material Selection

Component	Selected Material	Reason
Snowflake frame	Recycled Aluminium Alloy 6063-T5	Lightweight, corrosion-resistant and recyclable
Louvre blades	Powder-coated recycled aluminium alloy	Reduces actuator loading and maintenance
Mechanical linkage	Stainless Steel 316	High durability and corrosion resistance
Glazing	Double-glazed Low-E Glass	Reduces heat transfer while maintaining daylight
Sensor	LDR	Low-cost sunlight detection
Controller	Arduino Uno	Reliable and economical control
Actuator	SG90 Servo Motor (Prototype)	Suitable for kinetic movement

5.2.5 Scalability and Commercialisation Potential

The proposed Snowflake Kinetic Louvre Façade demonstrates strong scalability due to its modular configuration. Each snowflake unit functions independently, allowing application across different building sizes, orientations and façade conditions.

The system can be commercialised as a modular adaptive façade solution for new and existing commercial buildings through manufacturing, installation and maintenance services.

Scalability and Business Opportunities

Opportunity	Potential Benefit
Modular façade manufacturing	Production and sale of prefabricated snowflake modules
Installation and commissioning	Engineering and construction service opportunities
Maintenance contracts	Long-term inspection and replacement services
Retrofit applications	Upgrading existing commercial buildings

Smart-building integration	Future connection with building energy management systems
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The modular design reduces manufacturing complexity and allows gradual implementation, enabling the system to be scaled according to project requirements and available investment.

Overall, the Snowflake Kinetic Louvre Facade provides a balance between initial investment and long-term economic benefits. With an estimated **8.35-year payback period**, **11.97% annual ROI** and a service life exceeding **25 years**, the proposed system demonstrates potential as a financially viable adaptive façade solution for tropical commercial high-rise buildings.

5.3 Sustainability & Environmental Impact

The proposed Snowflake Kinetic Louvre Facade supports sustainable building development by reducing operational energy consumption, lowering carbon emissions and promoting efficient material usage. Through adaptive solar control, the system reduces solar heat gain before it enters the building, decreasing the cooling demand of commercial high-rise buildings in tropical climates.

5.3.1 Carbon Footprint Reduction and Energy Saving

In tropical regions, air-conditioning represents a significant portion of commercial building energy consumption due to high solar exposure and outdoor temperatures. The proposed kinetic façade reduces the amount of direct solar radiation entering the building by adjusting the louvre angle according to sunlight intensity.

Based on the estimated energy analysis in Section 5.2, the proposed system can reduce approximately **270,000 kWh/year of HVAC electricity consumption**. Assuming an electricity emission factor of approximately **0.585 kg CO₂-e/kWh**, the estimated annual carbon emission reduction is:

$$\begin{aligned}
 \text{Annual CO}_2 \text{ Reduction} &= \text{Annual Energy Saving} \times \text{Emission Factor} \\
 &= 270,000 \text{ kWh/year} \times 0.585 \text{ kg CO}_2\text{-e/kWh} \\
 &= 157,950 \text{ kg CO}_2\text{-e/year}
 \end{aligned}$$

Therefore, the proposed façade has the potential to reduce approximately **158 tonnes of CO₂-e annually** through reduced cooling-energy consumption.

5.3.2 Environmental Impact of Material Selection

Recycled aluminium is selected for the frame and louvre blades to reduce dependence on virgin resources and support material recycling. The lightweight aluminium components also reduce the mechanical load on the kinetic system.

The modular design supports design-for-disassembly, allowing individual louvres, sensors, actuators and other components to be replaced without removing the entire façade. This reduces replacement waste and can extend the useful life of the system.

The PV strips are integrated only into the fixed snowflake branches, preventing additional weight from being placed on the moving louvre assembly and reducing the mechanical demand on the actuators.

5.3.3 Life Cycle Assessment (LCA) Consideration

A complete Life Cycle Assessment (LCA) should be conducted during future full-scale implementation to evaluate environmental impacts throughout the façade lifecycle, including material production, transportation, installation, operation, maintenance and end-of-life disposal.

The proposed system is expected to provide environmental benefits throughout its lifecycle due to:

- Reduced operational energy consumption through adaptive solar shading.
- Use of recyclable aluminium materials.
- Reduced replacement waste through modular maintenance.
- Extended façade lifespan through corrosion-resistant materials.

Although the kinetic system requires additional materials and electronic components compared with static shading devices, the long-term operational energy savings are expected to offset the higher initial embodied impact.

5.3.4 Contribution to Sustainable Development Goals (SDGs)

The proposed Snowflake Kinetic Louvre Facade contributes to several United Nations Sustainable Development Goals (SDGs).

Contribution of Proposed Facade to SDGs

SDG	Contribution
SDG 7: Affordable and Clean Energy	Reduces building energy consumption through passive cooling strategies
SDG 9: Industry, Innovation and Infrastructure	Promotes smart façade technology and innovative building systems
SDG 11: Sustainable Cities and Communities	Supports energy-efficient and climate-responsive urban buildings
SDG 13: Climate Action	Reduces operational carbon emissions through lower HVAC demand

6.0 Implementation Strategy & Feasibility

6.1 Business Justification

6.1.1 Energy Cost Savings

HVAC accounts for up to 70% of energy use in Malaysia's commercial sector. (Samuji et al., 2026). High-rise office buildings in Malaysia use approximately 45.9% of their total building energy for cooling purposes (Lau et al., 2016). The facade slashes operational costs (OpEx) through three mechanisms:

- Solar Thermal Rejection:** An infrared-reflective (IR) powder coat ($SRI > 75$) is estimated to reflect a proportion of direct solar radiation at high shading angles, supporting the targeted reduction in cooling demand. The actual HVAC reduction will require validation through thermal simulation and prototype testing.
- TNB Peak Demand Shaving:** By deploying during peak afternoon hours, the skin flattens power spikes, eliminating costly Tenaga Nasional Berhad (TNB) Maximum Demand penalty charges.
- Active Light-Shelf Daylighting:** The opaque, tilted louvres bounce overhead sunlight onto the interior ceiling plate, providing glare-free ambient light (300–500+ lux) that allows perimeter electric lighting to stay off during the day.

6.1.2 Potential government incentives or support

The project aligns directly with Malaysia's National Energy Transition Roadmap (NETR) and qualifies for major government-backed financial frameworks. Under the Green Investment Tax Allowance (GITA) administered by MIDA and MGTC, qualifying green building assets receive a 100% Investment Tax Allowance on qualifying CapEx (MGTC, 2024), yielding a direct tax saving of RM 482,400 on the initial RM2,010,000 investment. Furthermore, debt financing can be optimized through Bank Negara Malaysia's Low Carbon Transition Facility (LCTF) which caps financing interest at a maximum of 5.0% p.a. for up to RM10 million or the Green Technology Financing Scheme (GTFS) which provides an interest rate subsidy of up to 2.0% p.a. off commercial bank loans (MGTC, 2021). These mechanisms significantly lower capital costs and de-risk the development for financial stakeholders."

6.2 Project Phases and Planning

6.2.1 Timeline

Project execution is structured across a 24-month timeline divided into five sequential phases. This schedule is engineered to manage technical and financial risks by subjecting the kinetic drive mechanisms to lab cycle testing and on-site podium validation before committing capital to full mass production. The timeline is shown in Figure 6.2.1.

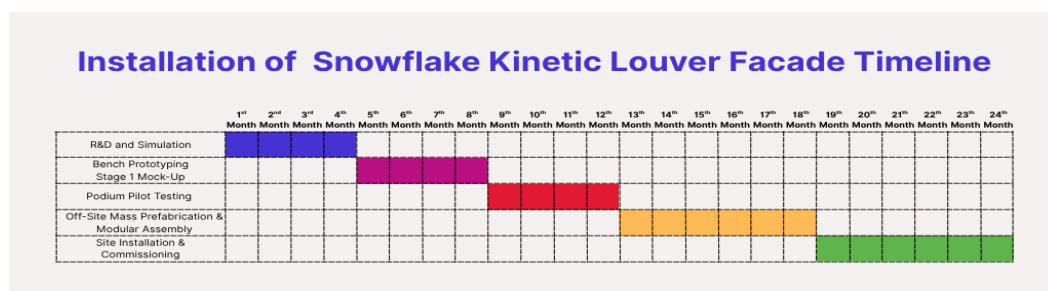


Figure 6.2.1: Gantt Chart of The Installation of Snowflake Kinetic Louvre Facade

6.2.2 Required Partners/Stakeholder

1. **Real Estate Developer & Lead Architect:** Overlook overall master planning, aesthetic sign-offs, project financing and integration with the primary curtain wall.
2. **Specialist Facade Contractor:** Handles extruded aluminum alloy fabrication, architectural powder coating application (Qualicoat Class 2 / PVDF finish), structural anchors and site installation.
3. **Mechatronics & Automation Integrator:** Supplies central servo drives, pre-tensioned Bowden-tendon assemblies, solar tracking sensors, microcontrollers and BMS communication interfaces.
4. **M&E & Structural Engineering Consultants:** Validates structural wind-load compliance under Malaysian Code of Practice (MS 1553) and recalculates reduced HVAC equipment capacity for CapEx downsizing.

6.2.3 Resources and Manpower

- a) **Lead Mechatronics Engineer (1.0 FTE):** Drives kinetic mechanics, Bowden-tendon tuning and laboratory fatigue testing.
- b) **Site Installation Lead / Riggers (1.0 FTE):** Oversees ground assembly, podium pilot mounting and high-rise crane operations.
- c) **Facade Structural Engineer (0.5 FTE):** Manages MS 1553 wind loading compliance, structural anchoring and facade sign-off.
- d) **Embedded Systems Developer (0.5 FTE):** Programs microcontrollers, sensor arrays and BMS network integration (Modbus/BACnet).
- e) **Sustainability & Grant Specialist (0.25 FTE):** Handles MIDA/MGTC GITA tax allowance filings, MyHIJAU registration and GBI points.
- f) **Software & Lab Rigs:** Ansys Fluent (CFD heat modeling), ANSYS Mechanical (FEA wind load) and an automated 100,000-cycle tendon fatigue test rig.
- g) **Fabrication & Pilot Equipment:** 5-axis CNC aluminum milling, PVDF high-SRI coating line and a 6m x 4m podium test frame with indoor temp/lux sensors.

6.2.4 Pilot Roadmap

Figure 6.2.4 shows the pilot roadmap of this project. To protect capital expenditure and ensure financial prudence, the pilot roadmap establishes a progressive stage-gate pipeline that verifies performance at each step before committing major funding to mass production.

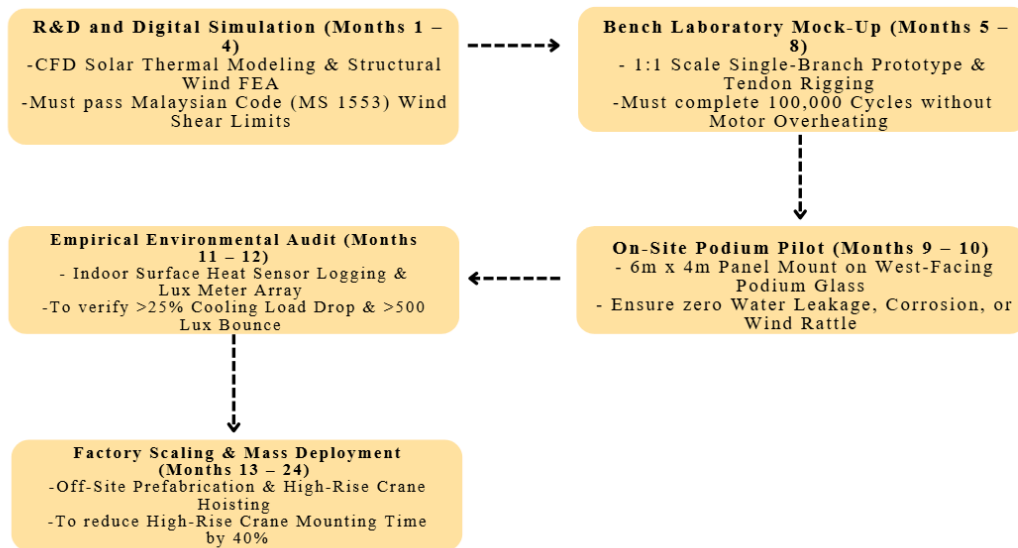


Figure 6.2.4: Pilot Roadmap

6.3 Risk Identification and Mitigation Strategies

Category	Risk	Mitigation Strategy
Technical	Servo/linkage wear or misalignment from repeated actuation	Low-friction shared-shaft linkage per blade pair; metal-gear servos (MG90) for durability
	LDR misreads under diffuse light or partial shading	Multi-point LDR array with averaging logic and a control deadband
	Wind loading causes vibration or damage at rotation axis	Static outer frame bears structural load; auto-lock Louvres flat above wind threshold
Performance	Real-world cooling reduction falls short of $\geq 25\%$ target	Iterative prototyping and OTTV-based thermal validation before full deployment
	Over-shading cuts daylighting, raising lighting load	Daylight-priority control logic during low-intensity periods
Financial	Per-unit sensor/servo cost limits scalability	Shared microcontroller per module cluster; low-cost off-the-shelf parts
Schedule	Prototype testing exceeds competition timeline	Validate single module first; build on existing working prototype
Market	Developers hesitant to adopt unproven kinetic systems	Position as modular, retrofit-friendly add-on, not full facade replacement

7.0 Conclusion

Static facades in Malaysia's tropical commercial buildings force an unavoidable trade-off: over-shading sacrifices daylight while under-shading drives HVAC systems to peak capacity, at a time when space cooling already accounts for up to 70% of commercial energy use and most facades operate with almost no thermal buffer left under the OTTV code limit.

The Snowflake Kinetic Louvre Facade addresses this through a biomimetic, modular system that separates structure from motion. A fixed hexagonal frame provides stability and wind resistance while lightweight trapezoidal Louvres rotate in real time to track the sun, filtering direct heat while preserving diffuse daylight. This design translates a proven natural principle of localized adaptive movement into a practical building envelope solution.

Key Benefits

Environmentally, the system targets a 25% reduction in cooling load and supports Malaysia's broader low-carbon building goals. Socially, occupants gain more consistent thermal comfort, reduced glare and stable daylight availability without manual intervention. Technically, isolating movement to the internal Louvres rather than the entire panel reduces mechanical complexity, wear and maintenance demands compared with conventional kinetic facades, while remaining compatible with retrofit installation.

Trade-offs

The system carries a higher upfront investment than static shading, estimated at RM2.01 million gross (RM1.5276 million net of the RM482,400 GITA tax saving) for a 3,000 m² installation, with a payback period of approximately 8.35 years on the net investment. Structural performance depends on the outer frame adequately absorbing wind loads so the rotating Louvres are not directly exposed to failure. Component availability, particularly servos, sensors and aluminium extrusions, must remain stable to support both prototype development and eventual full-scale deployment.

Feasibility and Impact

With a service life exceeding 25 years, government-backed incentives such as GITA and GTFS and a modular structure suited to phased retrofit, the Snowflake Kinetic Louvre Facade presents a technically sound and commercially viable path to lowering cooling demand in tropical commercial buildings.

As Malaysia's building stock faces rising cooling demand and tightening energy codes, adaptive facades like this one represent a necessary shift from static resistance to intelligent response, offering a scalable model for how tropical architecture can work with the sun rather than against it.

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9.0 Appendix

Appendix A: Concept Visualization

The following images present AI-assisted concept visualizations of the Snowflake Kinetic Louvre Facade module, illustrating its intended visual appearance in both operating states described in Section 4.2. These renderings are design concept aids and do not represent validated engineering drawings or physical prototypes; detailed dimensional specifications are provided in Section 4 and Appendix D.

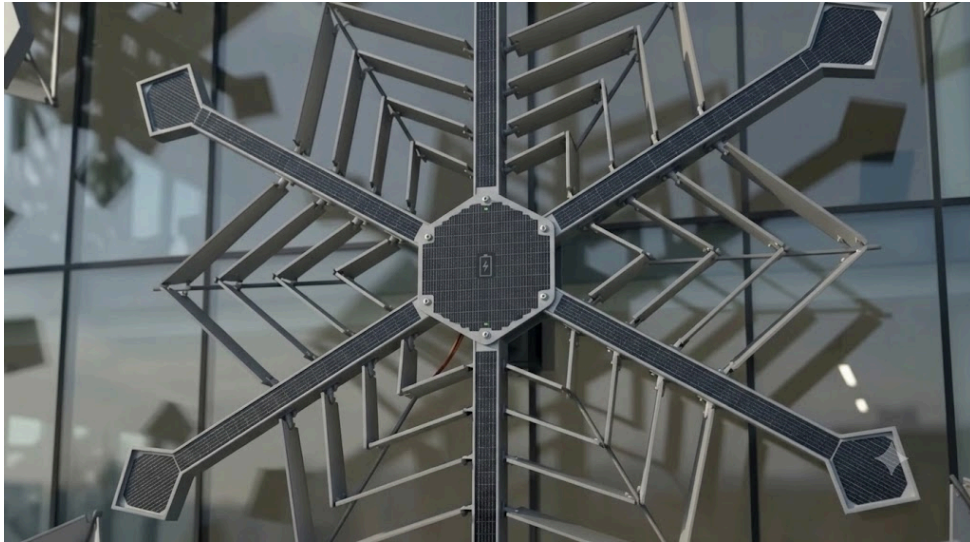


Figure A.1: AI-assisted concept visualization of the Snowflake Kinetic Louvre Facade in the open state (0°, low solar intensity), maximizing daylight transmission.

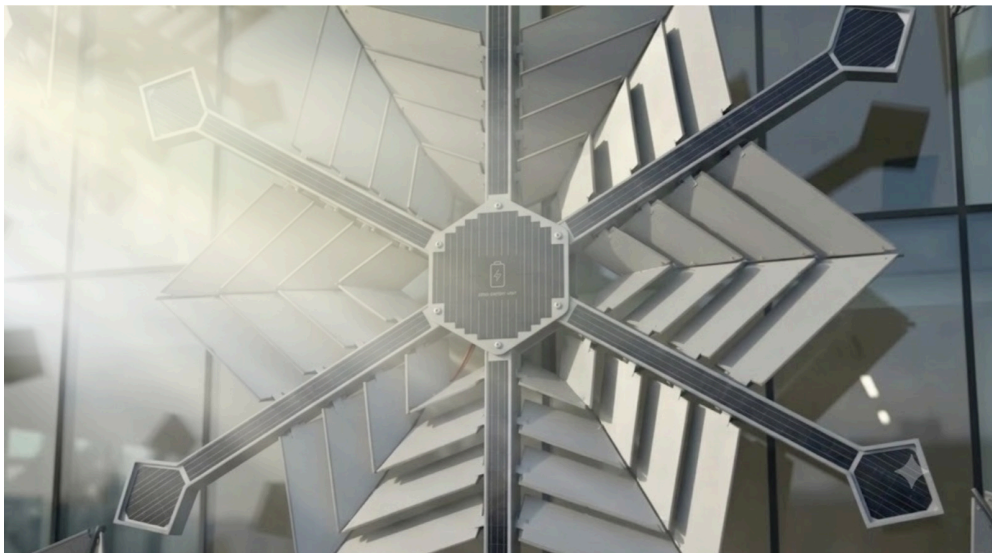


Figure A.2: AI-assisted concept visualization of the Snowflake Kinetic Louvre Facade in the shaded state (45°, high solar intensity), reducing direct solar heat gain.

Appendix B: Energy Saving Calculation

Building Parameters

Parameter	Value
Total building floor area	10,000 m ²
Glazed façade area	4,000 m ²
Kinetic façade coverage	3,000 m ² (75% of glazing)
HVAC energy intensity	128–160 kWh/m ² /year
Average HVAC energy intensity	144 kWh/m ² /year
Electricity tariff	RM0.40/kWh

Annual HVAC Electricity Consumption

Average HVAC energy intensity:

$$= (128 + 160) \div 2$$

$$= 144 \text{ kWh/m}^2/\text{year}$$

Annual HVAC electricity consumption:

$$= 144 \text{ kWh/m}^2/\text{year} \times 10,000 \text{ m}^2$$

$$= 1,440,000 \text{ kWh/year}$$

Affected Cooling Energy

$$= 1,440,000 \text{ kWh/year} \times 75\%$$

$$= 1,080,000 \text{ kWh/year}$$

Annual HVAC Energy Saving

Assuming 25% cooling demand reduction:

$$= 1,080,000 \text{ kWh/year} \times 25\%$$

$$= 270,000 \text{ kWh/year}$$

Annual HVAC Cost Saving

$$= 270,000 \text{ kWh/year} \times \text{RM0.40/kWh}$$

$$= \text{RM108,000/year}$$

Façade Electricity Consumption

Servo motors:

$$750 \times (5 \text{ V} \times 0.20 \text{ A}) = 750 \text{ W}$$

LDR sensors:

$$750 \times 0.00625 \text{ W} = 4.69 \text{ W}$$

Arduino controller:

$$= 0.25 \text{ W}$$

Total power consumption:

$$750 + 4.69 + 0.25 = 754.94 \text{ W} = 0.75494 \text{ kW}$$

Assuming 8 operating hours/day:

Daily electricity consumption:

$$0.75494 \times 8 = 6.04 \text{ kWh/day}$$

Annual electricity consumption:

$$6.04 \times 365 = \mathbf{2,205 \text{ kWh/year}}$$

PV Capacity and Generation

Total PV capacity:

$$750 \times 10 \text{ W} = 7,500 \text{ W} = 7.5 \text{ kWp}$$

Using a PV yield of 3.5 kWh/kWp/day:

Annual PV generation:

$$7.5 \times 3.5 \times 365 = 9,581 \text{ kWh/year}$$

PV Electricity-Cost Saving

Annual façade electricity demand = 2,205 kWh/year

Electricity tariff = RM0.40/kWh

PV electricity-cost saving:

$$2,205 \times \text{RM}0.40 = \mathbf{\text{RM}882/\text{year}}$$

Appendix C: Preliminary Solar Heat-Gain and Cooling-Load Reduction Calculation

Purpose of the Analysis

A preliminary solar heat-gain analysis was conducted to evaluate the potential of the proposed Snowflake Kinetic Louvre Façade in achieving the project target of reducing cooling load by at least 25% while maintaining useful natural daylight transmission.

The analysis estimates the reduction in solar heat gain through the 3,000 m² protected glazing area when the kinetic louvre system operates at the maximum designed rotation angle of 45°. This calculation serves as a preliminary engineering screening assessment to demonstrate the feasibility of the proposed concept and does not represent a validated whole-building energy simulation.

The solar heat gain through glazing is estimated using:

$$\mathbf{Q = SHGC \times I \times A}$$

Where:

Q = Solar heat gain through glazing (W)

SHGC = Solar Heat Gain Coefficient of glazing

I = Incident solar irradiance on façade (W/m²)

A = Protected glazing area (m²)

The SHGC value represents the glazing performance without external shading. The effect of the proposed Snowflake Kinetic Louvre Facade is evaluated separately as an additional external solar-control layer.

Input Parameters and Assumptions

The parameters used for the preliminary assessment are summarised in Table B.1.

Parameter	Symbol	Value	Unit
Protected glazing area	A	3,000	m ²
Glazing Solar Heat Gain Coefficient	SHGC	0.40	—
Peak façade solar irradiance	I_{peak}	700	W/m ²
Direct solar radiation fraction	f_{direct}	0.65	—
Diffuse solar radiation fraction	f_{diffuse}	0.35	—
Direct radiation intercepted by louvres	B_{direct}	80	%
Direct radiation transmitted	T_{direct}	20	%
Diffuse radiation transmitted through louver gaps	T_{diffuse}	70	%
Maximum louver rotation angle	θ	45	°
Solar contribution to perimeter-zone cooling load	f_{solar}	40	%
Required cooling-load reduction target	—	25	%

The SHGC value of 0.40 was selected as a representative preliminary value based on commercially available solar-control double-glazing systems. Manufacturer data show that solar-control glazing products can have g-values around 0.40, depending on the glass type and insulating-glass configuration.

Baseline Solar Heat Gain Without Kinetic Shading

The baseline condition represents the protected glazing without the proposed kinetic louver shading system.

The baseline solar heat gain is calculated using:

$$Q_{\text{baseline}} = \text{SHGC} \times I_{\text{peak}} \times A$$

Substituting the assumed values:

$$Q_{\text{baseline}} = 0.40 \times 700 \text{ W/m}^2 \times 3,000 \text{ m}^2$$

$$Q_{\text{baseline}} = 840,000 \text{ W}$$

Therefore:

$$Q_{\text{baseline}} = 840 \text{ kW}$$

The estimated solar heat gain through the protected glazing area is approximately 840 kW under the assumed peak solar condition.

Separation of Solar Radiation Components

The peak solar irradiance is separated into direct and diffuse components.

Direct Solar Irradiance:

$$I_{\text{direct}} = I_{\text{peak}} \times f_{\text{direct}}$$

$$I_{\text{direct}} = 700 \text{ W/m}^2 \times 0.65$$

$$I_{\text{direct}} = 455 \text{ W/m}^2$$

Diffuse Solar Irradiance

$$I_{\text{diffuse}} = I_{\text{peak}} \times f_{\text{diffuse}}$$

$$I_{\text{diffuse}} = 700 \text{ W/m}^2 \times 0.35$$

$$I_{\text{diffuse}} = 245 \text{ W/m}^2$$

Verification:

$$I_{\text{direct}} + I_{\text{diffuse}} = 455 + 245$$

$$I_{\text{total}} = 700 \text{ W/m}^2$$

Estimated Solar Radiation Transmission Through Louvre System

At the maximum louvre rotation angle of 45°, the trapezoidal louvre configuration is designed to significantly reduce direct solar penetration while allowing partial diffuse daylight transmission.

Based on preliminary geometric assessment, the louvre system is assumed to intercept approximately 80% of direct solar radiation, resulting in 20% direct radiation transmission.

Transmitted Direct Radiation

$$I_{\text{direct,transmitted}} = I_{\text{direct}} \times T_{\text{direct}}$$

$$I_{\text{direct,transmitted}} = 455 \text{ W/m}^2 \times 0.20$$

$$I_{\text{direct,transmitted}} = 91 \text{ W/m}^2$$

Transmitted Diffuse Radiation

To maintain useful daylight availability, approximately 70% of diffuse radiation is assumed to pass through the openings between louvre blades.

$$I_{\text{diffuse,transmitted}} = I_{\text{diffuse}} \times T_{\text{diffuse}}$$

$$I_{\text{diffuse,transmitted}} = 245 \text{ W/m}^2 \times 0.70$$

$$I_{\text{diffuse,transmitted}} = 171.5 \text{ W/m}^2$$

The total transmitted irradiance through the kinetic louvre system is:

$$I_{\text{transmitted}} = I_{\text{direct,transmitted}} + I_{\text{diffuse,transmitted}}$$

$$I_{\text{transmitted}} = 91 + 171.5$$

$$I_{\text{transmitted}} = 262.5 \text{ W/m}^2$$

Solar Heat Gain With Snowflake Kinetic Louvre Facade

The solar heat gain after applying the kinetic louvre shading system is estimated using:

$$Q_{\text{shaded}} = \text{SHGC} \times I_{\text{transmitted}} \times A$$

Substituting the calculated values:

$$Q_{\text{shaded}} = 0.40 \times 262.5 \text{ W/m}^2 \times 3,000 \text{ m}^2$$

$$Q_{\text{shaded}} = 315,000 \text{ W}$$

The proposed kinetic louvre system reduces estimated solar heat gain from 840 kW to approximately 315 kW under the assumed peak solar condition.

Estimated Solar Heat-Gain Reduction

The solar heat-gain reduction is calculated using:

$$\text{Solar Heat-Gain Reduction (\%)} = [(Q_{\text{baseline}} - Q_{\text{shaded}}) \div Q_{\text{baseline}}] \times 100\%$$

Substituting the calculated values:

$$\text{Solar Heat-Gain Reduction (\%)} = [(840 - 315) \div 840] \times 100\%$$

$$\text{Solar Heat-Gain Reduction (\%)} = 62.5\%$$

The preliminary analysis indicates that the proposed façade can potentially reduce peak solar heat gain through the protected glazing by approximately 62.5%.

Preliminary Cooling-Load Reduction Estimation

Solar heat gain represents only one component of the total cooling load of a commercial building. Other cooling-load components include:

- Heat gain from occupants
- Heat generated by electrical equipment
- Artificial lighting loads
- Outdoor air ventilation loads
- Heat transfer through walls, roofs and other building-envelope components

For highly glazed perimeter zones in tropical commercial buildings, solar heat gain is assumed to contribute approximately 40% of the total cooling load.

The estimated cooling-load reduction is calculated as:

Cooling-Load Reduction (%)

= Solar Heat-Gain Reduction × Solar Contribution to Cooling Load

= 62.5% × 40%

= **25.0%.**

For subsequent economic and environmental assessments, a conservative cooling-load reduction value of 25% is adopted to account for variations in façade orientation, weather conditions, building operation patterns and real-world performance limitations.

Summary of Results

Table B.2: Summary of Preliminary Solar Heat-Gain and Cooling-Load Results

Performance Parameter	Estimated Result
Protected glazing area	3,000 m ²
Baseline solar heat gain	840 kW
Solar heat gain after louvre shading	315 kW
Solar heat-gain reduction	62.5%
Assumed solar contribution to cooling load	40%
Estimated cooling-load reduction	25.0%
Project target	25.0%

Appendix D: Prototype Control Logic and Geometric Feasibility Verification

Purpose of this Appendix

This appendix presents two preliminary feasibility assessments supporting Sections 4.2 and 4.3: (1) an Arduino control logic simulation demonstrating the sensing-to-actuation process described in Section 4.2 and (2) a conceptual geometric assessment evaluating whether the proposed trapezoidal louvre arrangement provides sufficient clearance for 0°–45° rotation. Both serve as preliminary engineering verification prior to physical bench prototyping scheduled in Months 5–8 (Section 6.2.3).

D.1 Prototype Control Logic (Arduino Simulation)

To validate the sensing-to-actuation control logic prior to physical hardware integration, the core louvre control algorithm was developed and simulated in the Arduino IDE. The logic continuously reads real-time LDR light intensity, maps it proportionally to a louvre rotation angle between 0° (fully open, low solar intensity) and 45° (fully shaded, high solar intensity) and drives the servo motor accordingly, forming the closed-loop process described in Section 4.2.

```
cpp
#include <Servo.h>
```

```

Servo louvreServo;           // controls louvre rotation
const int ldrPin = A0;       // LDR sensor input
const int servoPin = 9;      // servo output

const int lightThresholdLow = 300; // low solar intensity
const int lightThresholdHigh = 800; // high solar intensity

void setup() {
  louvreServo.attach(servoPin);
  Serial.begin(9600);
}

void loop() {
  int lightLevel = analogRead(ldrPin); // read LDR (0-1023)

  int targetAngle;
  if (lightLevel < lightThresholdLow) {
    targetAngle = 0;           // fully open - low sun
  } else if (lightLevel > lightThresholdHigh) {
    targetAngle = 45;          // fully shaded - high sun
  } else {
    // linear interpolation between thresholds
    targetAngle = map(lightLevel, lightThresholdLow, lightThresholdHigh, 0, 45);
  }

  louvreServo.write(targetAngle);
  Serial.print("Light: "); Serial.print(lightLevel);
  Serial.print(" | Louvre Angle: "); Serial.println(targetAngle);

  delay(200); // control loop update interval
}

```

Table D.1: Control Logic Parameters

Parameter	Value
Light sensor input range	0–1023 (analog)
Low-intensity threshold	300
High-intensity threshold	800
Louvre angle range	0°–45°
Control loop update interval	200 ms

This simulation demonstrates the feasibility of the proposed closed-loop control logic—sensing, angle mapping and actuation, prior to full hardware validation during the Bench Laboratory Mock-Up phase.

D.2 Preliminary Geometric Design Verification

A schematic layout of the snowflake module was developed to verify that the trapezoidal louvre arrangement allows rotation from 0° to 45° without obvious overlap between adjacent branches. Adequate spacing was incorporated during the design stage to provide clearance for the rotating blades and linkage mechanism, as illustrated in Figure C.2.1. Detailed mechanical interference analysis will be carried out during future CAD assembly and physical prototype testing in the Bench Laboratory Mock-Up phase (Months 5–8).



Figure D.2.1: Schematic of a single snowflake branch showing louvre rotation states — 0° (fully open, maximum daylight) and 45° (fully shaded, gap retained between adjacent blades).

Table D.3: Preliminary Geometric Assessment

Parameter	Value
Maximum louvre rotation angle	45°
Maximum projected blade width at 45° tilt ($\approx 220 \text{ mm nominal} \times \cos 45^\circ$)	155.4mm
Available arc spacing between adjacent branches	1570 mm
Preliminary assessment	Sufficient spacing incorporated for conceptual design

This preliminary geometric check indicates the proposed module dimensions are mechanically feasible at the conceptual stage. Precise clearances will be further validated through CAD modelling and physical prototyping in Months 5–8.